Introduction to Molecular Simulation

From Bits to Bonds — Computing Matter, One Atom at a Time

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What is Simulation? — 60s Primer

Click to watch: What is Simulation?

What is Simulation?

youtube.com/watch?v=XPE7eaR_XxM

Outline — Roadmap

What is Simulation? — Control Over Numbers on a Chip

- **Simulation** = computable mirror of reality

Chip = Lab

Transistors $\to$ arithmetic $\to$ models $\to$ atoms $\to$ materials $\to$ life

CPU/GPU
chip

$\downarrow$ model

Simulation
$F = ma\,, U(\mathbf{r})$

$\downarrow$ measure

Observables
$P,\ \rho,\ D,\ \Delta G$

What is Simulation? — Control Over Numbers on a Chip

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- **Control** $\rightarrow$ numbers you **ingrain** in silicon

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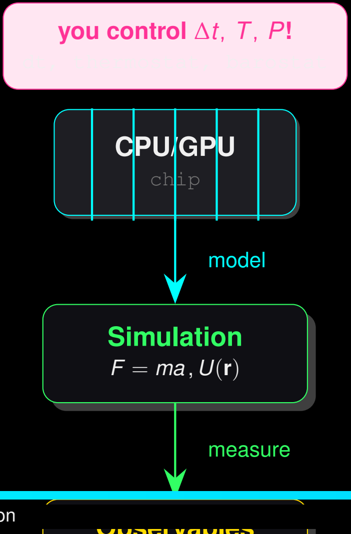

What is Simulation? — Control Over Numbers on a Chip

- **Simulation** = computable mirror of reality
- **Control** $\longrightarrow$ numbers you **ingrain** in silicon
- Chip = deterministic universe you design

CPU/GPU
chip

model

Simulation
$F = ma\,,\ U(\mathbf{r})$

measure

Observables
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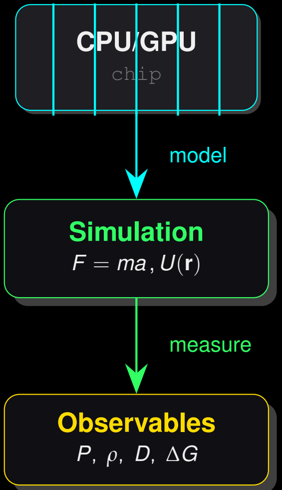
10^9 **steps** $\longrightarrow$ 1

Chip = Lab

Transistors $\longrightarrow$ arithmetic $\longrightarrow$ models $\longrightarrow$ atoms $\longrightarrow$ materials $\longrightarrow$ life

What is Simulation? — Control Over Numbers on a Chip

- **Simulation** = computable mirror of reality
- **Control** $\to$ numbers you **ingrain** in silicon
- Chip = deterministic universe you design
- Input: `numbers` $\to$ Output: `properties`



Chip = Lab

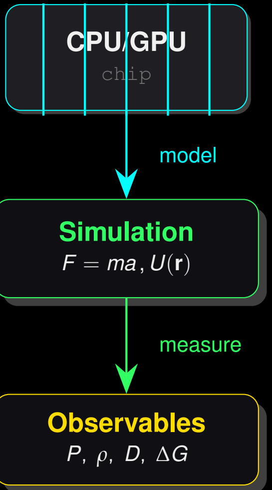
Transistors $\to$ arithmetic $\to$ models $\to$ atoms $\to$ materials $\to$ life

What is Simulation? — Control Over Numbers on a Chip

- **Simulation** = computable mirror of reality
- **Control** $\rightarrow$ numbers you **ingrain** in silicon
- Chip = deterministic universe you design
- Input: `numbers` $\rightarrow$ Output: `properties`
- **Why?** — experiment impossible / dangerous / expensive

Chip = Lab

Transistors $\rightarrow$ arithmetic $\rightarrow$ models $\rightarrow$ atoms $\rightarrow$ materials $\rightarrow$ life



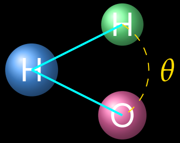
Interpretation — Simulation

Interpretation — keywords

Simulation · mimic reality in code · **control** numbers $\rightarrow$ control physics · **chip** = lab · **reproducible** · **cheap** *what-if* · **not** magic — model quality = result quality

Molecular Modeling — From Orbitals to Atoms

◇ **Orbitals** $\psi_{n\ell m} \to$ electron density



Wireframe $\leftrightarrow$ Ball-and-stick

$\leftrightarrow$ Space-filling

Molecular Modeling — From Orbitals to Atoms

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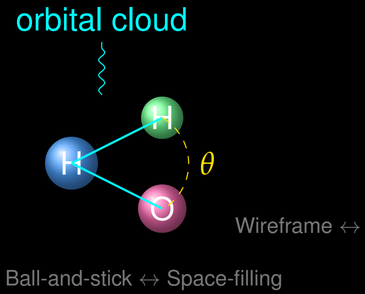
- Hydrogenic: $\hat{H}\psi = E\psi$, $\psi_{100} \propto e^{-r/a_0}$



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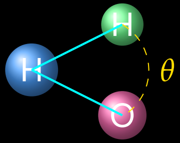

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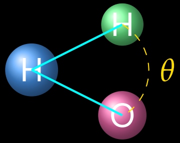

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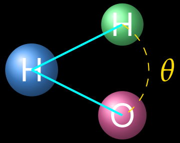
- Precision ladder: **hydrogenic** $\rightarrow$ DFT $\rightarrow$ force field



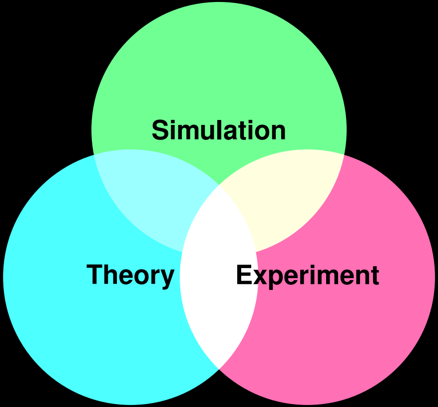
Wireframe $\leftrightarrow$ Ball-and-stick $\leftrightarrow$ Space-filling

Interpretation — Modeling

Interpretation — keywords

Modeling · choose resolution · **orbitals** = quantum truth · **atoms** = classical shortcut · **precision** costs · **ball-and-stick** = human eye model

Computational Experiments — The Third Pillar



- **Theory** $\rightarrow$ equations

Not a video game

You don't watch molecules — you **measure** them: $g(r)$, MSD, ΔG.

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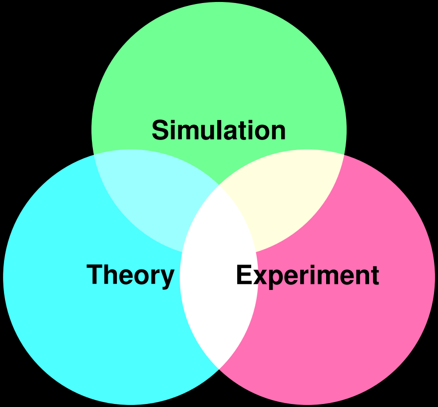

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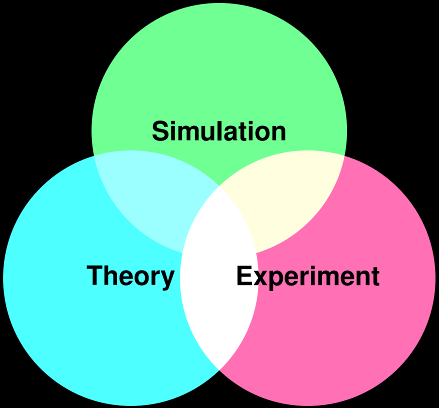

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- **Simulation** $\rightarrow$ **nature under your conditions**

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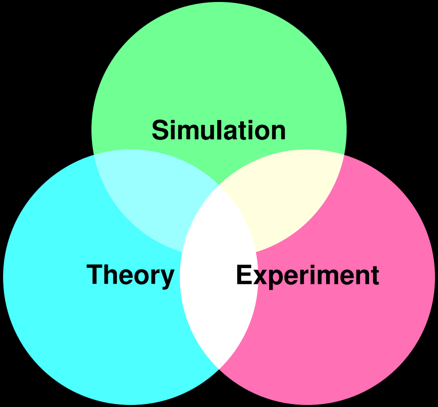

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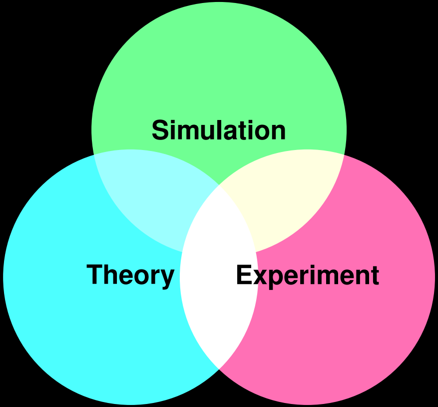

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- Falsifiable, reproducible

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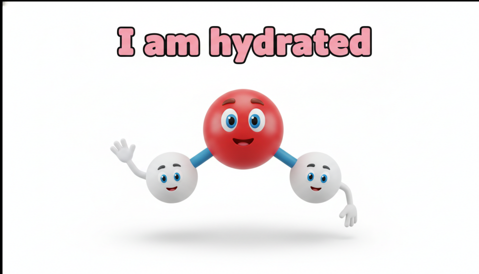
Molecular Representation — How Computers See Molecules

- **Ball-and-stick**: atoms=spheres, bonds=sticks



Meshy `nano-banana` — water waving

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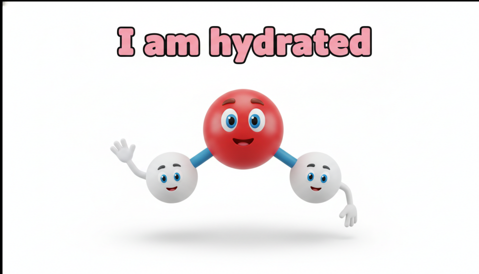

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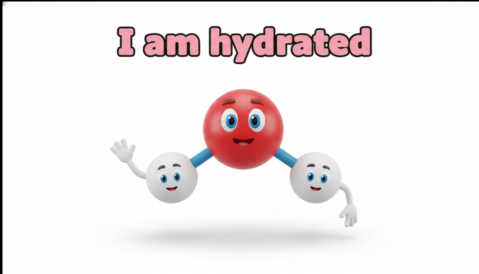

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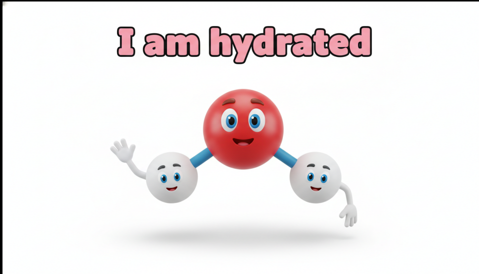

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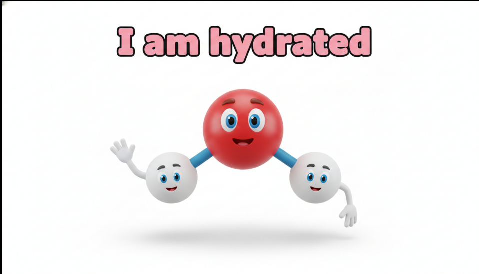
- All derived from `coordinates + bonds`



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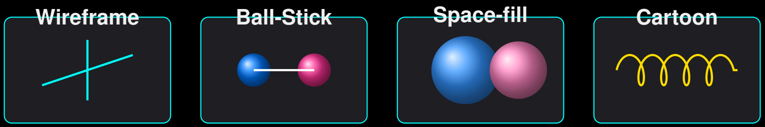
TikZ — Representation Zoo



Simulation Objectives — What Are We After?

- **Structure**: $g(r)$, SASA, clustering

Success = Question + Model + Sampling

Bad question + good model = noise
Good question + bad model = lie

Drug binding → Materials → Energy → B

Simulation Objectives — What Are We After?

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- **Design**: new molecule / material

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Applications — Where It Matters

- Pharma: binding ΔG, **AlphaFold+MD**

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Applications — Where It Matters

- Pharma: binding ΔG, **AlphaFold+MD**
- Materials: battery, polymer, catalyst
- Energy: CO2 capture, membranes
- Life: folding, crowding, rare events

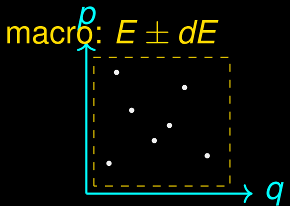
Interpretation — Objectives

Interpretation — keywords

Objectives · **structure** / **dynamics** / **thermodynamics** · **mechanism** over movie · **design** is the test · **applications** drive approximations

Microstate vs Macrostate

- **Microstate**: $\Gamma = \{\mathbf{r}^N, \mathbf{p}^N\}$ — everything



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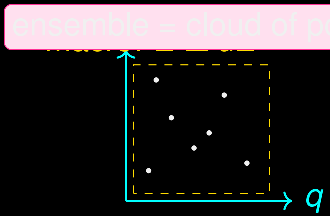

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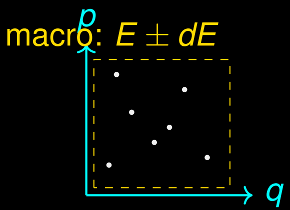

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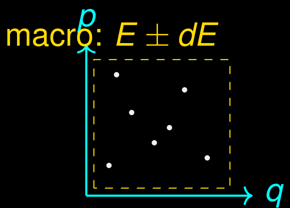
- Phase space: $6N$-dim, point = microstate



Interpretation — Micro/Macro

Interpretation — keywords

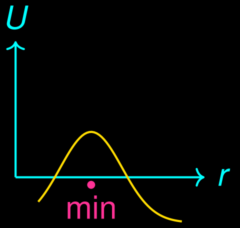
Micro · exact positions·momenta · **macro** · thermometer numbers · **phase space** · map of all micros · **one macro = many micros**

Energy — The Currency

$$E = K + U = \sum_i \frac{\mathbf{p}_i^2}{2m_i} + U(\mathbf{r}^N)$$

micro definition

- $U(\mathbf{r}^N)$ = PES — see §3



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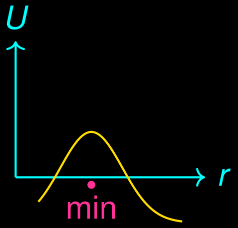
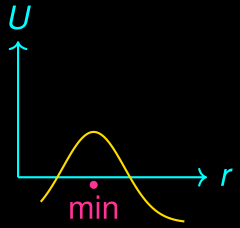


Energy — The Currency

$$E = K + U = \sum_i \frac{\mathbf{p}_i^2}{2m_i} + U(\mathbf{r}^N)$$ micro definition

- $U(\mathbf{r}^N)$ = PES — see §3
- Conserved in NVE: $dE/dt = 0$
- Fluctuates in NVT: $\langle(\Delta E)^2\rangle = k_B T^2 C_V$



Entropy — Counting

$$S = k_B \ln \Omega \qquad \text{and} \qquad S = -k_B \sum_i p_i \ln p_i$$

- ◇ Ω = # microstates for macro

$\boxed{\Omega = 1}$

$\downarrow S \uparrow$

$\boxed{\Omega = 10^{23}}$

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- ◇ More ways $\to$ more entropy $\to$ more likely

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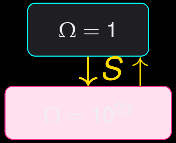
- Ω = # microstates for macro
- More ways $\rightarrow$ more entropy $\rightarrow$ more likely
- Disorder = **counting**, not mess

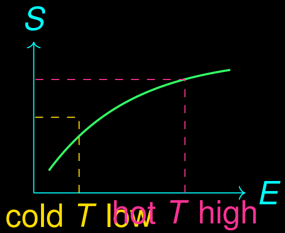


Temperature — Slope of Entropy

$$\frac{1}{T} = \left(\frac{\partial S}{\partial E}\right)_{N,V}$$

hot = small slope

- Cold: S steep vs E



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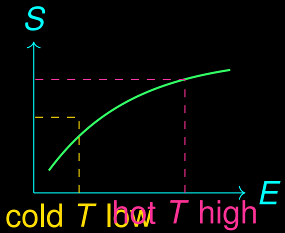

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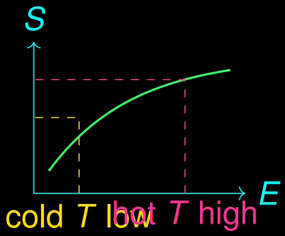
- Cold: S steep vs E
- Hot: S flat vs E
- T = **exchange rate** of $E \leftrightarrow$ S



Interpretation — Energy/Entropy/Temperature

Interpretation — keywords

Energy · conserved currency · **entropy** · count of ways · **temperature** · $dS\,/\,dE$ · **hot** = flat $S(E)$

Free Energy — What Actually Matters

$$F = U - TS \qquad G = H - TS \qquad \min F, G \Leftrightarrow \text{stable}$$

- U wants order, S wants disorder

Free Energy — What Actually Matters

$$F = U - TS \qquad G = H - TS \qquad \min F, G \Leftrightarrow \text{stable}$$

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- F = compromise at T

Free Energy — What Actually Matters

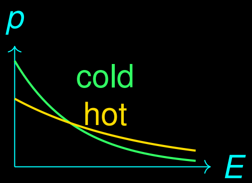
$$F = U - TS \qquad G = H - TS \qquad \min F, G \Leftrightarrow \text{stable}$$

- U wants order, S wants disorder
- F = compromise at T
- Spontaneous $\Delta F < 0$

Boltzmann Distribution

$$p_i = \frac{e^{-\beta E_i}}{Z}, \quad \beta = \frac{1}{k_B T}, \quad Z = \sum_i e^{-\beta E_i}$$

- Low $E \to$ high p



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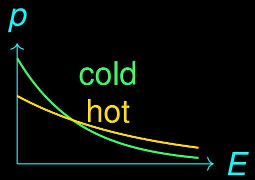
- High $T \rightarrow$ flatter p



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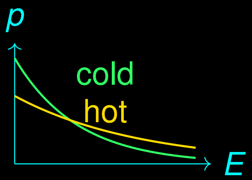
- High $T \rightarrow$ flatter p
- Z = partition function = normalization + **generator**



Partition Function

$$Z = \sum e^{-\beta E} \text{ (or } \int e^{-\beta H} d\Gamma) \qquad F = -k_B T \ln Z$$

- $Z \to F \to P$, S, C_V by derivatives

Partition Function

$$Z = \sum e^{-\beta E} \ (\text{or} \int e^{-\beta H} d\Gamma) \qquad F = -k_B T \ln Z$$

- $Z \to F \to P$, S, C_V by derivatives
- Never need absolute Z, need **ratios** $\to$ **sampling**

Interpretation — Free Energy/Boltzmann

Interpretation — keywords

free energy · balance U vs TS · **Boltzmann** · low E wins · Z · sum that rules them all · **cold** = picky, **hot** = tolerant

NVE, NVT, NPT — Choose Your Lab

NVE
fix E
micro/canonical/isobaric

thermostat
$\leftrightarrow$

NVT
fix T
micro/canonical/isobaric

barostat
$\leftrightarrow$

NPT
fix T, P
micro/canonical/isobaric

- **NVE**: isolated, E conserved $\rightarrow$ MD default

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- **NVE**: isolated, E conserved $\rightarrow$ MD default
- **NVT**: heat bath, T fixed $\rightarrow$ biology lab

NVE, NVT, NPT — Choose Your Lab

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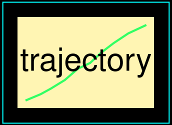
- **NVE**: isolated, E conserved $\rightarrow$ MD default
- **NVT**: heat bath, T fixed $\rightarrow$ biology lab
- **NPT**: P+ T fixed $\rightarrow$ real world

Ergodicity & Ensemble Average

$$\langle A \rangle = \sum_i A_i p_i \approx \frac{1}{T} \int_0^T A(t)\, dt \quad \text{if ergodic}$$

- **Ergodic**: trajectory visits all with correct weight



ensemble (shaded)

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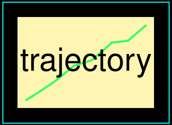

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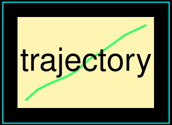
- Broken $\rightarrow$ trapped, **rare events** matter
- Solution: **enhanced sampling** (§9)



ensemble (shaded)

Interpretation — Ensembles

Interpretation — keywords

ensembles · what you fix · **NVE/NVT/NPT** · lab choice · **ergodic** · time = ensemble · **broken** $\rightarrow$ sample smarter

Atomic Coordinates — The $3N$ Truth

$$\mathbf{R} = \{\mathbf{r}_1, \ldots, \mathbf{r}_N\}, \quad \mathbf{r}_i = (x_i, y_i, z_i) \in \mathbb{R}^3$$

- $3N$ numbers = **complete** classical description

```
ATOM 1 C 0.0 0.0 0.0
```

```
ATOM 2 O 1.2 0.0 0.0
```

```
ATOM 3 H -0.4 0.9 0.0
```

array

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- **Rigorous**: coordinates **are** the state

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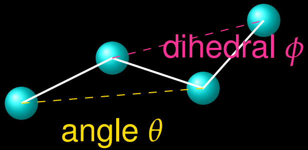
Bonds, Angles, Dihedrals — Topology

- ◇ **Bond**: $r_{ij} = |\mathbf{r}_j - \mathbf{r}_i|$ · 2 atoms



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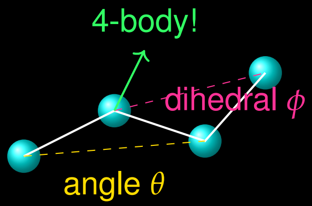

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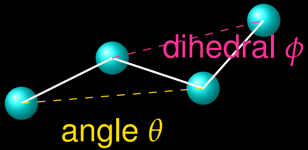

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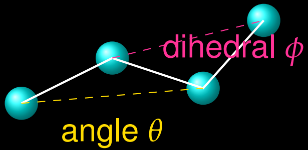
- Topology = **graph** (bonds = edges)



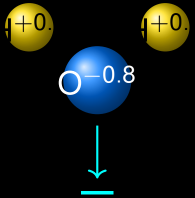
Interpretation — Coordinates/Topology

Interpretation — keywords

coordinates · $3N$ numbers = molecule · **bond** · distance · **angle** · 3 atoms · **dihedral** · 4 atoms = flexibility · **topology** · graph

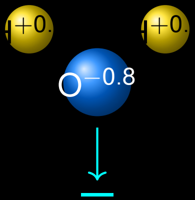
Masses & Partial Charges

◇ $m_i \to$ dynamics: $\mathbf{F} = m\mathbf{a}$



Masses & Partial Charges

- $m_i \rightarrow$ dynamics: $\mathbf{F} = m\mathbf{a}$
- $q_i \rightarrow$ electrostatics: $U_{\text{Coul}} = \frac{q_i q_j}{4\pi\epsilon_0 r_{ij}}$



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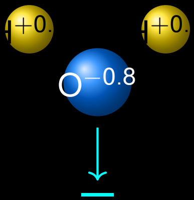

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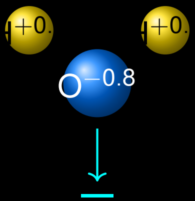
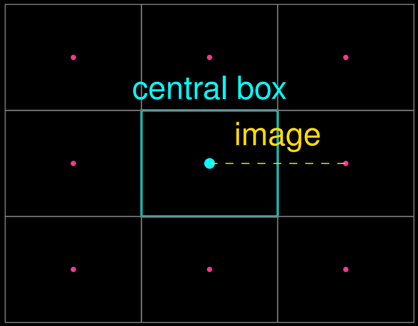
- Rigorous: charges are **parameters**, not observables



Simulation Box & Periodic Boundary Conditions

◇ Box: $\mathbf{L} = (L_x, L_y, L_z)$, triclinic: $\mathbf{h}$



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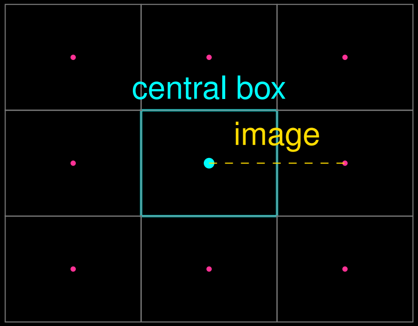

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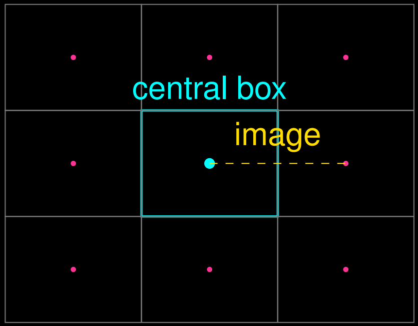

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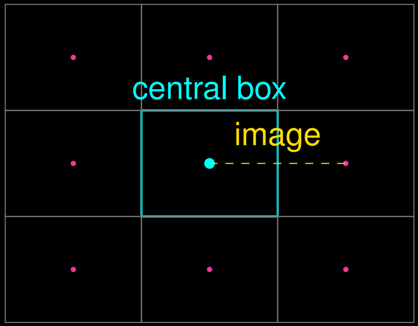
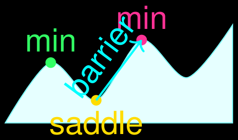
- Cutoff $r_c < L/2$



Potential Energy Surface — The Mountain Range

$$U(\mathbf{R}) : \mathbb{R}^{3N} \to \mathbb{R} \qquad \mathbf{F}_i = -\nabla_{\mathbf{r}_i} U$$

- 3N-dim surface $\to$ 2D cartoon is **projection**



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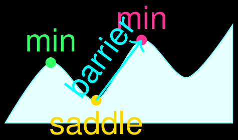

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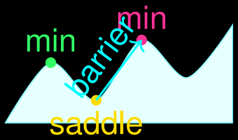

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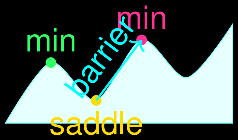
- ◇ $3N$-dim surface $\to$ 2D cartoon is **projection**
- ◇ Minima = stable structures
- ◇ Saddle = transition state
- ◇ Rigorous: **all** physics in U



Force Fields — U Made Computable

$$U = \underbrace{U_{\text{bond}} + U_{\text{angle}} + U_{\text{dihed}}}_{\text{bonded}} + \underbrace{U_{\text{LJ}} + U_{\text{Coul}}}_{\text{nonbonded}}$$

- Bond: $k_b(r - r_0)^2$ · spring

FF = Expensive Physics Compressed

10^5 DFT points $\to U_{\text{FF}} \to 10^9$ steps

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- Bond: $k_b(r - r_0)^2$ · spring
- LJ: $4\epsilon[(\sigma/r)^{12} - (\sigma/r)^6]$
- Many: AMBER, CHARMM, OPLS, ReaxFF, **ML potentials**

FF = Expensive Physics Compressed

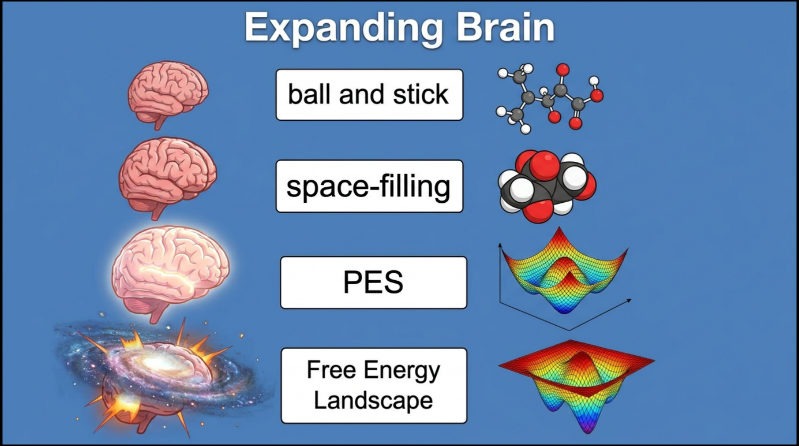
10^5 DFT points $\to U_{\text{FF}} \to 10^9$ steps

Interpretation — PES/FF

Interpretation — keywords

PES · energy landscape · **force** = slope · **force field** · cheap U · **bonded + nonbonded** · **parameterized** $\rightarrow$ not universal

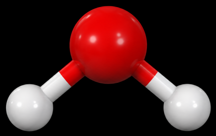
Meme — Expanding Brain of Representation



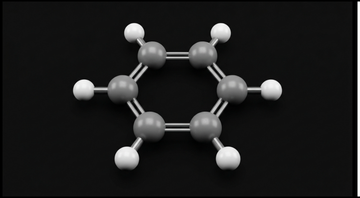
Expanding Brain:

ball-and-stick → PES → Free Energy Landscape

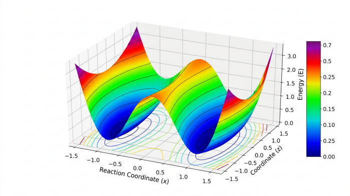
Realistic Figures — Ball-and-Stick & PES



H_2O: ∠ 104.5°, ball-and-stick

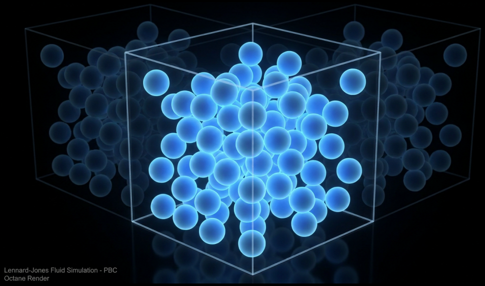


Benzene: 6-membered ring

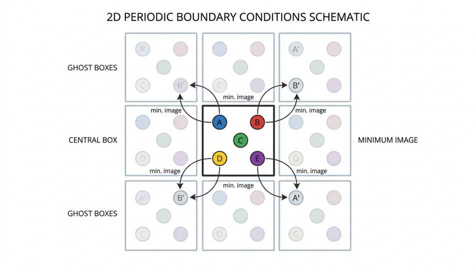


PES: double well + saddle

Realistic Figures — Simulation Box & PBC



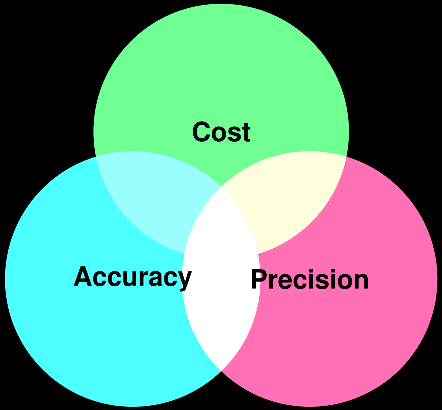
Cubic box, LJ fluid, PBC ghost images



Minimum image convention

Accuracy — What Does *Correct* Mean?

- **Accuracy** = closeness to **reality**



You can be precise and still wrong.

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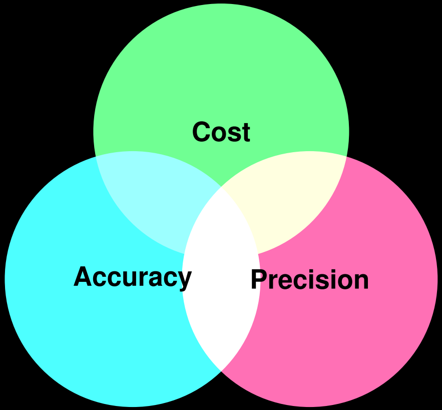

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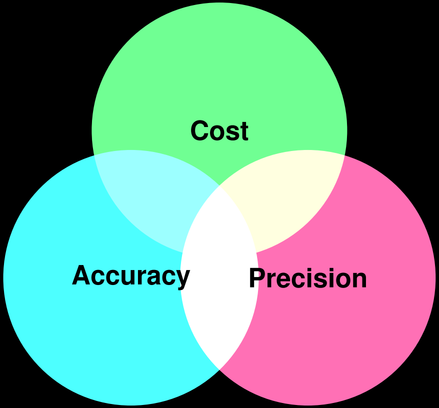

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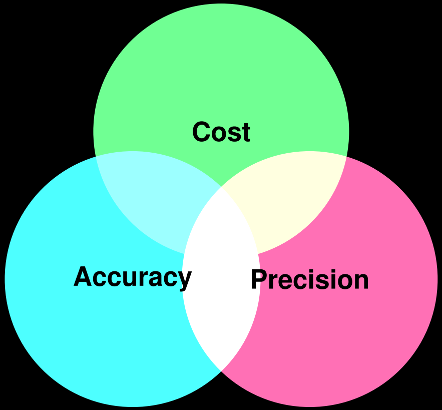
- **Errors** stack: model + numerics + sampling



You can

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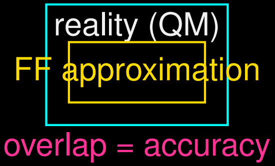
Force-Field Approximations

◇ Fixed topology → no bond breaking (except ReaxFF)



Force-Field Approximations

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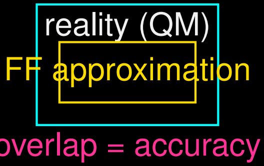

Force-Field Approximations

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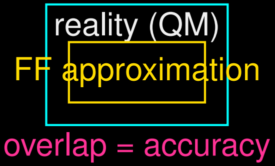

Force-Field Approximations

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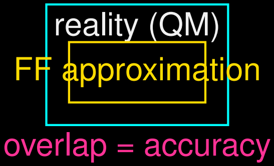
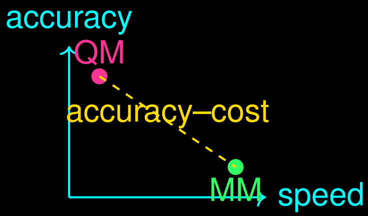
- Point charges → no polarization (unless Amoeba/Drude)
- Harmonic bonds → no anharmonicity at large r



Quantum vs Classical — Trade-off

- **QM** (DFT): $\hat{H}\psi = E\psi$, electrons explicit, $N \sim 10^2$, $t \sim$ ps



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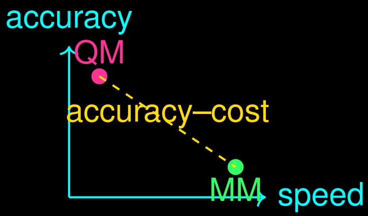
- **MM** (FF): $F = ma$, atoms, $N \sim 10^6$, $t \sim \mu$s



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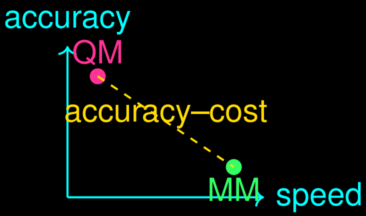
- **Scale**: QM accuracy $\rightarrow$ MM speed



Meme — Drake: Classical vs ML

Drake: rejecting fixed charges → approving ML potentials

Parameterization — Fitting Reality

$$\min_{\{k_b, r_0, \epsilon, \sigma, q\}} \sum \left| U_{\mathrm{FF}} - U_{\mathrm{QM/exp}} \right|^2$$

- Fit to QM energies/forces or experimental ρ, ΔH_{vap}

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- Validate outside training set!

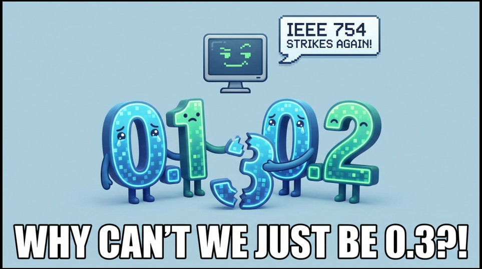
Numerical Errors — Computers Are Discrete

- Real $\in \mathbb{R} \to$ float $\in$ IEEE 754 (e.g., binary64: 1-11-52)

```
1 bit | 11 exp | 52 mantissa
```

$\downarrow$round

$0.1_{10} = 0.000110011\ldots_2$



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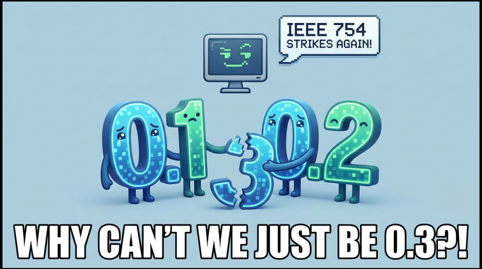

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- Error $\epsilon_{\text{mach}} \sim 10^{-16}$ (double)

```
1 bit | 11 exp | 52 mantissa
```

$\downarrow$round

$0.1_{10} = 0.000110011\ldots_2$

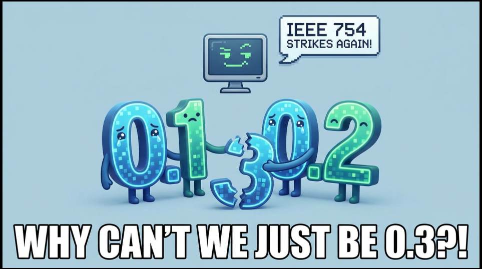
Numerical Errors — Computers Are Discrete

- Real $\in \mathbb{R} \to$ float $\in$ IEEE 754 (e.g., binary64: 1-11-52)
- $0.1 + 0.2 \neq 0.3$ (binary)
- Error $\epsilon_{\text{mach}} \sim 10^{-16}$ (double)
- Accumulates: 10^9 steps $\times$ dt error

```
1 bit | 11 exp | 52 mantissa
```

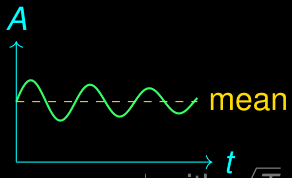
$\downarrow$round

$0.1_{10} = 0.000110011..._2$



Sampling Errors — Statistics

- Finite time $T \rightarrow$ error $\propto 1/\sqrt{T/\tau}$



noise $\rightarrow$ error $\downarrow$ with $\sqrt{T}$

Sampling Errors — Statistics

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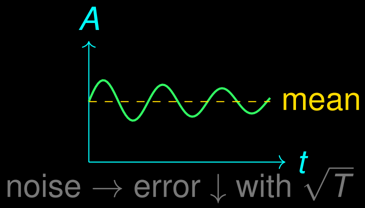
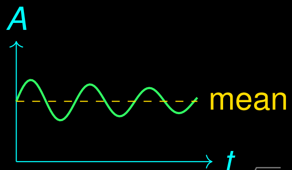


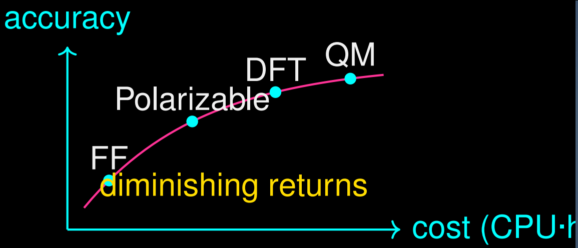
Sampling Errors — Statistics

- Finite time $T \rightarrow$ error $\propto 1/\sqrt{T/\tau}$
- τ = correlation time (ps to ms)
- **Block averaging** $\rightarrow$ error bars



noise $\rightarrow$ error $\downarrow$ with $\sqrt{T}$

Accuracy–Cost Trade-off



Two Buttons: accuracy vs cost

Interpretation — Accuracy

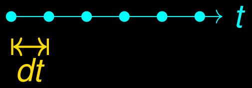
Interpretation — keywords

accuracy · vs. Nature · **FF** = biggest error · **QM vs MM** · speed · **IEEE 754** · $0.1 + 0.2 \neq 0.3$ · **sampling** · $1/\sqrt{T}$ · **trade-off** · pay for truth

The Timestep *dt* — Heartbeat of MD

$$\mathbf{r}(t + dt) = \mathbf{r}(t) + \mathbf{v}dt + \frac{1}{2}\mathbf{a}dt^2 + \cdots$$

- *dt* must resolve fastest vibration (bond $\sim$ 10 fs)



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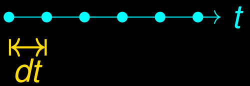
$$\mathbf{r}(t+dt) = \mathbf{r}(t) + \mathbf{v}dt + \tfrac{1}{2}\mathbf{a}dt^2 + \cdots$$

- *dt* must resolve fastest vibration (bond ~ 10 fs) SHAKE $\to$ $dt = 2$ fs!
- Rule: $dt \lesssim 1$ fs (flexible), 2 fs (constrained H)



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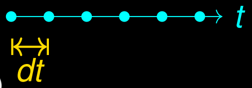
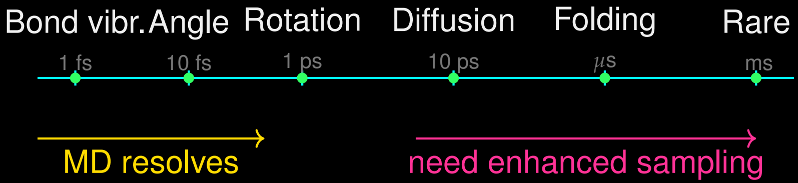
- *dt* must resolve fastest vibration (bond $\sim$ 10 fs)
- Rule: $dt \lesssim$ 1 fs (flexible), 2 fs (constrained H)
- Too large $\rightarrow$ **explosion** (energy drift)



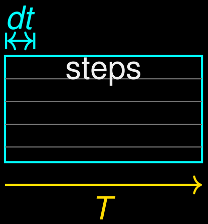
Hierarchy of Time



◇ **fs** $\rightarrow$ bonds, **ps** $\rightarrow$ water, **ns** $\rightarrow$ sidechains, μ**s** $\rightarrow$ folding, **ms–s** $\rightarrow$ rare

Simulation Length — $T = N_{\text{steps}} \times dt$

- $N_{\text{steps}} = T \,/\, dt$



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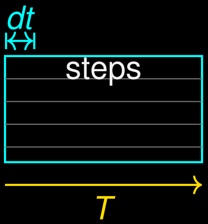
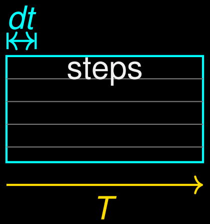


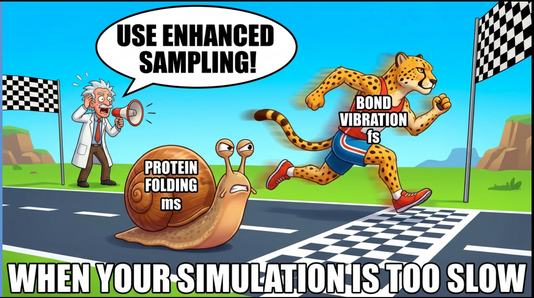
Simulation Length — $T = N_{\text{steps}} \times dt$

- $N_{\text{steps}} = T\,/\,dt$
- 100 ns at 2 fs $\rightarrow$ 50M steps
- Cost: 10^9 force calls = GPU-days



Meme — Timescales (This is Fine + Race)

This is Fine: dt=5 fs in production



Snail (ms) vs Cheetah (fs)

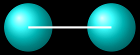
Interpretation — Timescales

Interpretation — keywords

dt · 1–2 fs · **hierarchy** · fs$\to$ms · $T = Ndt$ · steps = cost · **rare** · need tricks · **clock** vs **biology** mismatch

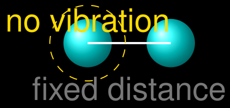
Static Structures — Frozen Snapshots

◇ **Static**: single **R**, no T, no motion



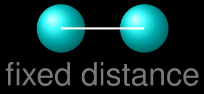
Static Structures — Frozen Snapshots

- **Static**: single **R**, no T, no motion
- Where you look: **crystal**, **PDB**, **DFT optimum**



Static Structures — Frozen Snapshots

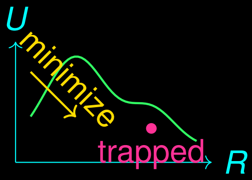
- **Static**: single **R**, no T, no motion
- Where you look: **crystal**, **PDB**, **DFT optimum**
- Question: *what is stable?*



Energy Minimization

$$\min_{\mathbf{R}} U(\mathbf{R}) \quad \Rightarrow \quad \mathbf{F} = -\nabla U = 0,\ \text{Hessian} > 0$$

- Steepest descent / CG / L-BFGS



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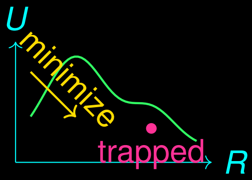

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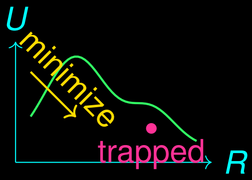
- No T, no entropy



Geometry Optimization & DFT

- **Geom. opt.**: MM minimization (FF)

DFT $\approx$ Exact for U, Still Static

No temperature $\rightarrow$ no entropy $\rightarrow$ free energy missing

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- **DFT**: QM minimization, *ab initio* U
- Cost: DFT $\sim 10^3 \times$ MM

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Geometry Optimization & DFT

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- **DFT**: QM minimization, *ab initio* U
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DFT $\approx$ Exact for U, Still Static

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The Paradox — Why Proteins Fold Fast

- Protein: 10^2 aa, $3^{100} \sim 10^{47}$ conformations

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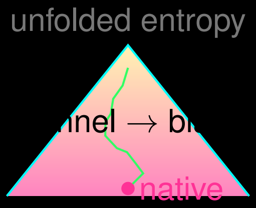
The Paradox — Why Proteins Fold Fast

- Protein: 10^2 aa, $3^{100} \sim 10^{47}$ conformations
- Random search: 10^{27} yr
- Reality: ms–s $\rightarrow$ **Levinthal paradox**
- Resolution: **funnel**, not flat — downhill + guides

Funnel TikZ



Thermodynamic Stability — F, Not U

$$\Delta G = \Delta H - T\Delta S \qquad p \propto e^{-\beta G}$$

- Min $U \neq$ min F at $T > 0$

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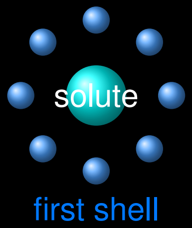
$$\Delta G = \Delta H - T\Delta S \qquad p \propto e^{-\beta G}$$

- Min $U \neq$ min F at $T > 0$
- Entropy pushes to **broad** basins
- Solvent: **hydrophobic** = entropy of water

Solvent Effects

- Explicit water: 10^4 molecules, PBC



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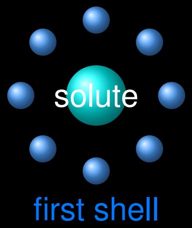
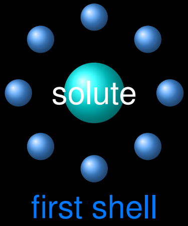


Solvent Effects

- Explicit water: 10^4 molecules, PBC
- Implicit: GB/SA, continuum
- Rigorous: solvent = **free energy**, not just screening



Interpretation — Static vs Dynamic

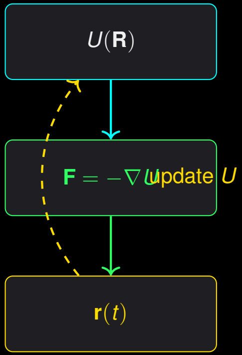
Interpretation — keywords

static · 0 K · min U · **Levinthal** · 10^{47} vs ms · **funnel** solves · **stability** = min F · **solvent** = entropy · **dynamic** = MD needed

Newton's Equations — The Engine

$$m_i \ddot{\mathbf{r}}_i = \mathbf{F}_i = -\nabla_{\mathbf{r}_i} U(\mathbf{R}) \qquad \mathbf{v}_i = \dot{\mathbf{r}}_i$$

- $U \rightarrow$ force $\rightarrow$ acceleration $\rightarrow$ move



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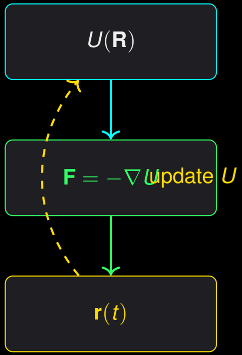

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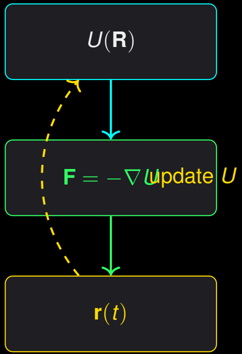
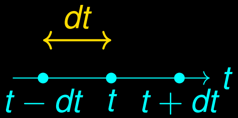
- Need: **positions**, **velocities**, **forces**
- Deterministic if U exact



Numerical Integration — Verlet

$$\mathbf{r}(t+dt) = 2\mathbf{r}(t) - \mathbf{r}(t-dt) + \frac{\mathbf{F}(t)}{m}dt^2 + O(dt^4)$$

- **Verlet**: time-reversible, symplectic $\rightarrow$ stable



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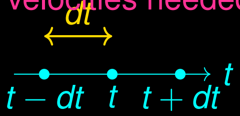
- **Errors**: $O(dt^4)$ per step, $O(dt^2)$ global

no velocities needed!



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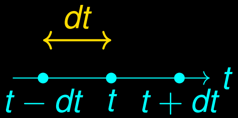
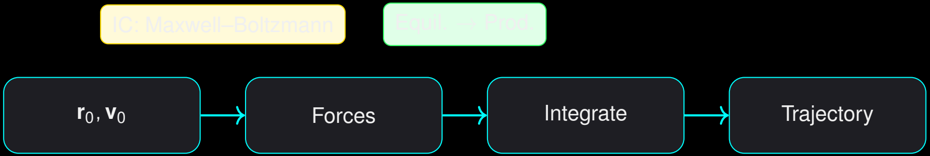
- **Errors**: $O(dt^4)$ per step, $O(dt^2)$ global
- SHAKE/RATTLE $\rightarrow$ constrain H-bonds $\rightarrow$ larger dt

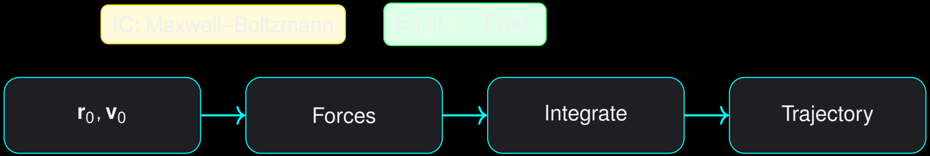


MD Workflow — From r_0 to Trajectory



- **Initial conditions**: positions (PDB) + velocities ($\propto \sqrt{T}$)

MD Workflow — From $\mathbf{r}_0$ to Trajectory



- **Initial conditions**: positions (PDB) + velocities ($\propto \sqrt{T}$)
- **Equilibration**: relax T, P, density plateau

MD Workflow — From r_0 to Trajectory

IC: Maxwell–Boltzmann

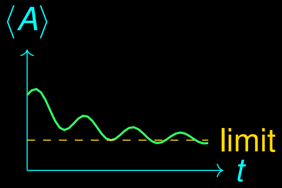
Equil. → Prod.

$\mathbf{r}_0, \mathbf{v}_0$ → Forces → Integrate → Trajectory

- **Initial conditions**: positions (PDB) + velocities ($\propto \sqrt{T}$)
- **Equilibration**: relax T, P, density plateau
- **Production**: N steps $\rightarrow$ **trajectory** + **sampling**

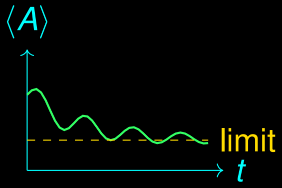
Convergence — When to Stop?

- **Convergence**: $\langle A \rangle$ stops drifting



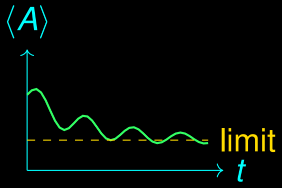
Convergence — When to Stop?

- **Convergence**: $\langle A \rangle$ stops drifting
- Check: $\langle T \rangle$, RMSD plateau, block error

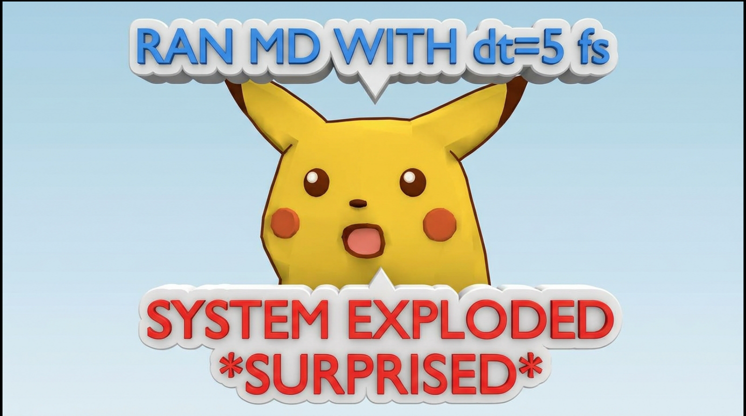


Convergence — When to Stop?

- **Convergence**: $\langle A \rangle$ stops drifting
- Check: $\langle T \rangle$, RMSD plateau, block error
- **Longer $\neq$ converged** if τ huge



Meme — Pikachu Surprise



When dt=5 fs and system explodes — *surprised*

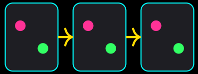
Interpretation — MD

Interpretation — keywords

Newton · $F = -\nabla U$ · **Verlet** · reversible · **IC** · Maxwell · **trajectory** · equil.$\rightarrow$prod. · **convergence** · plateau · **sampling** = statistics

Trajectory — The Movie

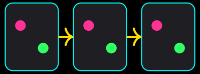
- **Trajectory**: $\{\mathbf{R}(t_0), \mathbf{R}(t_1), \ldots\}$ — DCD/XTC



frames $\rightarrow$ Δt

Trajectory — The Movie

- **Trajectory**: $\{\mathbf{R}(t_0), \mathbf{R}(t_1), \ldots\}$ — DCD/XTC
- Size: $N \times 3 \times N_{\text{frames}} \times 4$ bytes



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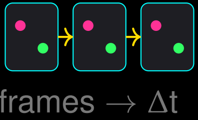
- Size: $N \times 3 \times N_{\text{frames}} \times 4$ bytes
- **Analysis** = compress movie $\rightarrow$ numbers



RMSD, RMSF, R_g

RMSD

$\mathrm{RMSD}(t) = \sqrt{\frac{1}{N}\sum_i |\mathbf{r}_i(t) - \mathbf{r}_i^{\mathrm{ref}}|^2}$
drift = motion

RMSF

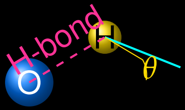
$\mathrm{RMSF}_i = \sqrt{\langle(\mathbf{r}_i - \langle\mathbf{r}_i\rangle)^2\rangle}$
flexibility per residue

$\mathbf{R}_g$

$R_g^2 = \frac{1}{N}\sum_i |\mathbf{r}_i - \mathbf{r}_{\mathrm{COM}}|^2$
compactness

H-Bonds, Distances, Angles

- **H-bond**: $d_{\mathrm{O}\cdots\mathrm{H}} < 3.5\text{Å}$, $\angle > 150°$



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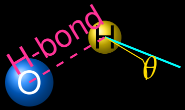
- Distances/angles $\rightarrow$ **order parameters**



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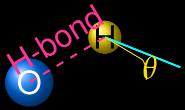
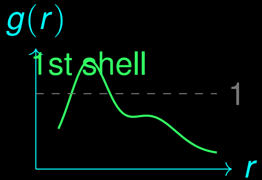
- Distances/angles $\rightarrow$ **order parameters**
- **Define** before measuring



Radial Distribution Function

$$g(r) = \frac{\langle \rho(r) \rangle}{\rho_0} \quad \text{— shell structure}$$

- First peak $\rightarrow$ coordination shell



Radial Distribution Function

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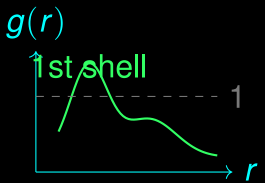
- $g(r) \rightarrow 1$ at long r



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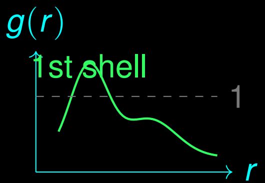
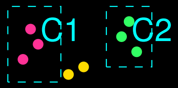
- Integrates to $n(r) = 4\pi\rho_0 \int_0^r g(s) s^2 ds$



Conformational Clustering

- Group frames by RMSD → **clusters**



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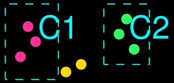
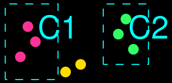


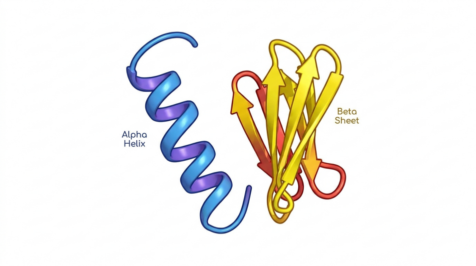
Conformational Clustering

- Group frames by RMSD $\rightarrow$ **clusters**
- Representative = medoid
- **Why?** $\rightarrow$ reduce 10^6 frames $\rightarrow$ 5 states



Visualization — See Before You Believe

- **VMD/PyMOL/OVITO/ChimeraX** $\rightarrow$ see artefacts



PyMOL-style cartoon: helices + sheets

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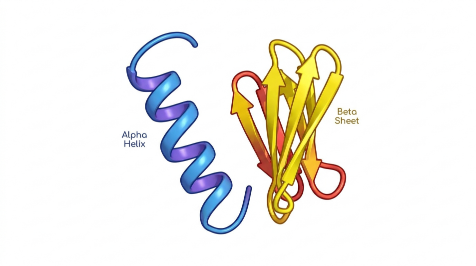

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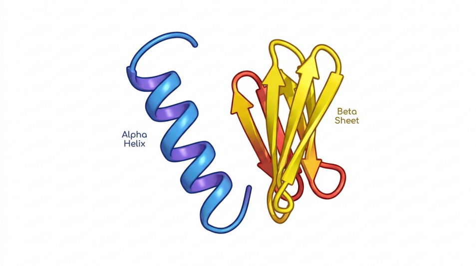
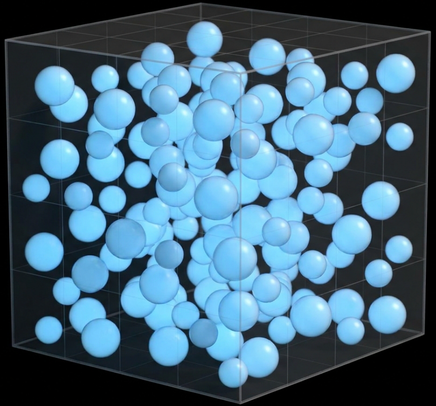
- **VMD/PyMOL/OVITO/ChimeraX** $\rightarrow$ see artefacts
- Color by **velocity**, **charge**, time
- **Titan** anecdote: 18,000 GPUs, 1 μs/day $\rightarrow$ still **need** analysis



PyMOL-style cartoon: helices + sheets

Realistic Figures — LJ Fluid Snapshot



Lennard-Jones Fluid, N~100 Ar Lennard-Jones fluid: what your trajectory actually looks like (OVITO view)

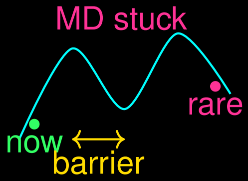
Interpretation — Analysis

Interpretation — keywords

trajectory · movie · **RMSD** · drift · **RMSF** · flex · R_g · compact · **H-bond** · geometry · $g(r)$ · shells · **clustering** · 5 states · **visualize** · trust

The Sampling Problem

- MD samples $\propto e^{-\beta U} \rightarrow$ rare events **never** seen



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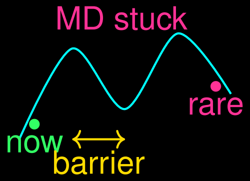
- Barrier $10\, k_B T \rightarrow$ wait $e^{10} \sim 20\text{k}\times$ longer



The Sampling Problem

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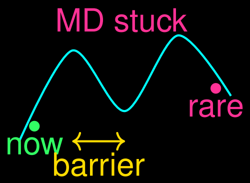
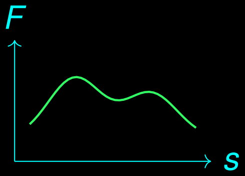
- **Biology lives in rare events**



Metadynamics — Fill the Well

$$V_{\text{bias}}(s, t) = \sum_{t'} w\, e^{-(s-s(t'))^2/2\sigma^2}$$

- Deposit Gaussians along CV s



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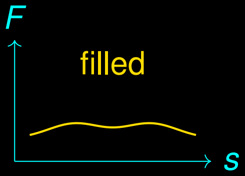

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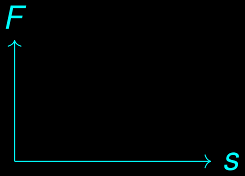
$$V_{\text{bias}}(s,t) = \sum_{t'} w\, e^{-(s-s(t'))^2/2\sigma^2}$$

- Deposit Gaussians along CV s
- Flood minima $\rightarrow$ escape
- **Free energy** = $-V_{\text{bias}}$ at convergence



Umbrella Sampling — Windows

- Harmonic umbrella per window: $w_i(s) = \frac{1}{2}k(s - s_i)^2$

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- Overlap windows $\rightarrow$ WHAM/MBAR

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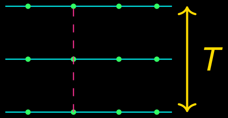
- Harmonic umbrella per window: $w_i(s) = \frac{1}{2}k(s - s_i)^2$
- Overlap windows $\rightarrow$ WHAM/MBAR
- Rigorous *PMF* via reweighting

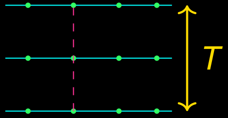
Replica Exchange & Accelerated MD

- **Replica exchange**: hot replicas $\rightarrow$ cold via swap $P \propto e^{\Delta\beta\Delta U}$



Replica Exchange & Accelerated MD

- **Replica exchange**: hot replicas $\rightarrow$ cold via swap $P \propto e^{\Delta\beta\Delta U}$
- **Accelerated MD**: boost $U \rightarrow U + \Delta V$ when $U <$ threshold



Potential of Mean Force

$$W(s) = -k_B T \ln P(s) \qquad P(s) = \langle \delta(s - s(\mathbf{R})) \rangle$$

- PMF = free energy **along** CV

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$$W(s) = -k_B T \ln P(s) \qquad P(s) = \langle \delta(s - s(\mathbf{R})) \rangle$$

- PMF = free energy **along** CV
- Minimum = stable, barrier = rate via Kramers

Interpretation — Enhanced Sampling

Interpretation — keywords

rare · barrier wins · **metadynamics** · fill wells · **umbrella** · windows $\rightarrow$ WHAM · **replica** · heat helps · **PMF** · $-kT \ln P$ · **free energy** = truth

Ecosystem — Who Does What

LAMMPS | GROMACS | OpenMM | NAMD | AMBER | CHARMM

engines

VMD | PyMOL | OVITO | ChimeraX

viewers

◇ **LAMMPS**: materials, all-atom, coarse-grained, **you built it**

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- **LAMMPS**: materials, all-atom, coarse-grained, **you built it**
- **GROMACS**: biomolecules, fast, PME
- **OpenMM**: GPU, Python API, custom forces

Choosing — Cheat Sheet

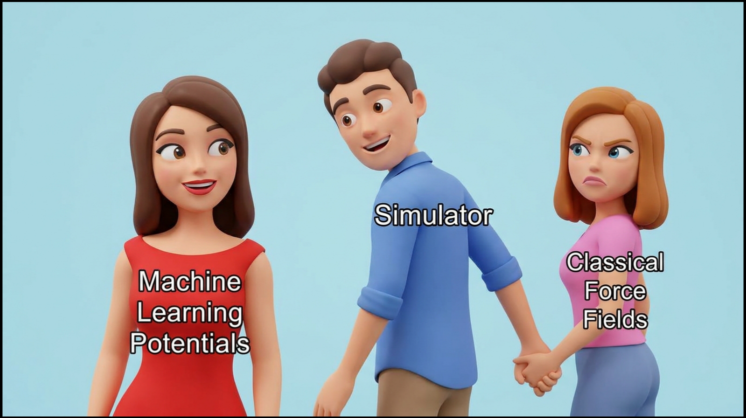
Pick by System

Metals/polymers $\to$ LAMMPS
Biomembrane/protein $\to$ GROMACS/AMBER
Custom $\to$ OpenMM
Visualize $\to$ VMD/OVITO

All Need

Force field **+++** **PBC** + **thermostat** + **analysis**

Meme — Distracted Simulator



Distracted Boyfriend: Simulator

$\rightarrow$ ML potentials (goodbye Classical FF)

Interpretation — Software

Interpretation — keywords

LAMMPS · materials · **GROMACS** · bio · **OpenMM** · GPU Python · **VMD/OVITO** · eyes · **all** · U+PBC+sampling

Installation — One Script, Both OS

```
# Linux ( Xeons 24c + A100 sm_80 )
./scripts/build-lammps.sh    #
↪  most+gpu+kokkos
# Windows PowerShell
.\scripts\build-lammps.ps1
# Check
./lammps/build/lmp -h | head
```

You Built 65 Packages

`lmp -h` shows GPU (mixed, sm_80) + KOKKOS (CUDA, AMPERE80) + 24 cores

Input Script — The Recipe

```
units real
atom_style full
read\_data water.data
pair\_style lj/cut/coul/long 10
kspace_style pppm 1e-4
fix 1 all nvt temp 300 300 100
timestep 2.0
run 50000
```

- **Box:** `region + create_box` or `read_data`

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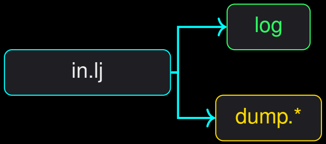
- **dt**: 1–2 fs

Execution — Log, Trajectory, Analysis

- `lmp -in in.lj -log log.lammps`



Execution — Log, Trajectory, Analysis

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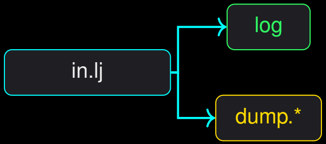
- **Log**: thermo (step, T, E, P)



Execution — Log, Trajectory, Analysis

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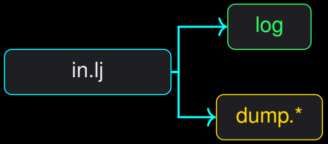

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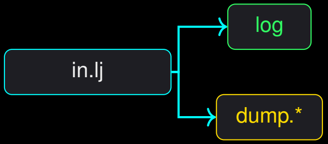
- **Analysis**: `compute rdf`, Python post



Interpretation — LAMMPS

Interpretation — keywords

install · `build-lammps.sh` · **script** · 5 lines → run · **log** · T, E, P · **dump** · movie · **analysis** · your science

Installation & Files

```
# Install (conda / apt)
conda install -c conda-forge gromacs
# Inputs
gmx pdb2gmx -f protein.pdb
gmx editconf -f conf.gro -o box.gro -c
  -d 1.0 -bt cubic
gmx solvate -cp box.gro -o solv.gro
```

- **Structure:** `.gro`/`.pdb`

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- **Structure**: `.gro`/`.pdb`
- **Topology**: `topol.top` (FF + bonds)
- **Parameters**: `.mdp` (dt, T, P)

GROMACS Workflow — Min $\to$ NVT $\to$ NPT $\to$ Prod

`em.mdp` $\to$ `nvt.mdp` $\to$ `npt.mdp` $\to$ `md.mdp`

Minimize $\to$ NVT $\to$ NPT $\to$ Production

- **EM**: min U, $F_{\text{max}} < 1000$ kJ/mol/nm

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Minimize → NVT → NPT → Production

- **EM**: min U, $F_{\text{max}} < 1000$ kJ/mol/nm
- **NVT**: V-rescale thermostat, $dt = 2$ fs, LINCS

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- **NVT**: V-rescale thermostat, $dt = 2$ fs, LINCS
- **NPT**: Parrinello-Rahman, $P = 1$ bar
- **Prod**: 50 ns $\rightarrow$ `traj.xtc` + `ener.edr`

Trajectory Analysis

```
gmx rms -s topol.tpr -f traj.xtc -o rmsd.xvg
gmx gyrate -s topol.tpr -f traj.xtc -o rg.xvg
gmx rdf -s topol.tpr -f traj.xtc -o rdf.xvg
```

- Same as LAMMPS: RMSD, R_g, $g(r)$, H-bonds

Trajectory Analysis

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- Same as LAMMPS: RMSD, R_g, $g(r)$, H-bonds
- Different file names, same physics

Interpretation — GROMACS

Interpretation — keywords

GROMACS · bio focus · **files** · gro/top/mdp · **workflow** · min→NVT→NPT→prod · **analysis** · same $g(r)$, RMSD · **PMEs** = fast Coulomb

Linux & Bash — Your Lab Bench

```bash
for d in */; do
  mpirun -np 24 lmp -in in.lj
  python analyse.py traj.xtc
done
```

- **Linux**: HPC speaks bash

Linux & Bash — Your Lab Bench

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- **Linux**: HPC speaks bash
- **Automation**: loops, slurm, ‘snakemake‘
- **Reproducible**: script beats clicking

Python Stack — NumPy → MDAnalysis

- **NumPy/SciPy**: arrays, FFT, linalg

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import MDAnalysis as mda
import numpy as np
u = mda.Universe("topol.tpr","traj.xtc")
for ts in u.trajectory:
    pos = u.atoms.positions
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- **RDKit**: cheminformatics

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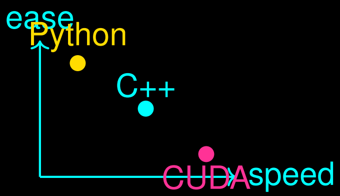
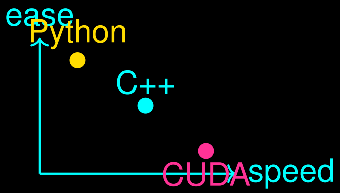
C/C++/Fortran/CUDA — Speed

◇ LAMMPS/GROMACS = **C++**/CUDA/Fortran



C/C++/Fortran/CUDA — Speed

- LAMMPS/GROMACS = **C++**/CUDA/Fortran
- You: Python for glue, **CUDA** for custom kernels



C/C++/Fortran/CUDA — Speed

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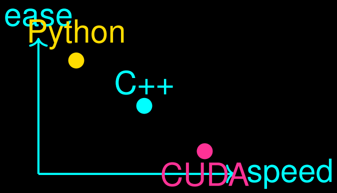
- You: Python for glue, **CUDA** for custom kernels
- `uv` → C++ → LAMMPS plugin



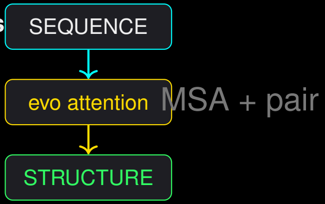
Interpretation — Programming

Interpretation — keywords

Bash · loop it · **Python** · NumPy/MDAnalysis · **C++/CUDA** · speed · **automation** · snakemake · **repro** = script

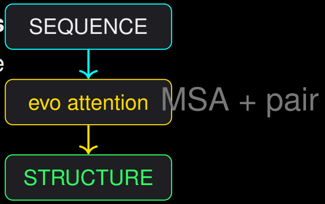
AlphaFold / OpenFold — Sequence → Structure

- Input: **sequence** → Output: **3D coordinates**



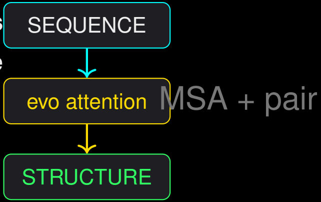
AlphaFold / OpenFold — Sequence → Structure

- Input: **sequence** → Output: **3D coordinates**
- Attention over MSA → distances → structure module



AlphaFold / OpenFold — Sequence → Structure

- Input: **sequence** → Output: **3D coordinates**
- Attention over MSA → distances → structure module
- **Accuracy**: CASP, lDDT > 90 for many



AlphaFold / OpenFold — Sequence → Structure

- Input: **sequence** → Output: **3D coordinates**
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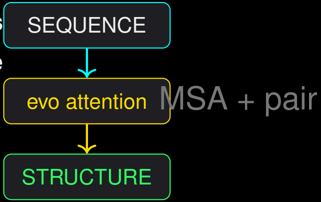
- **Accuracy**: CASP, IDDT > 90 for many
- **OpenFold** = open AlphaFold2



Prediction vs Simulation

AlphaFold

One **static** structure
No dynamics, no F, no ensemble

MD

Ensemble of structures
T, F, pathways, rates

AlphaFold = photo vs MD = movie

NVIDIA NIM & AI-Assisted Modeling

- **NIM**: containerized inference → AlphaFold3, OpenFold on GPUs

$$\text{AI predict} \xrightarrow{\text{seed}} \text{MD sample/fine-tune}$$

NVIDIA NIM & AI-Assisted Modeling

- **NIM**: containerized inference $\rightarrow$ AlphaFold3, OpenFold on GPUs
- **AI + MD**: AI gives start, MD refines + samples **free energy**

NVIDIA NIM & AI-Assisted Modeling

- **NIM**: containerized inference → AlphaFold3, OpenFold on GPUs
- **AI + MD**: AI gives start, MD refines + samples **free energy**
- Rigorous: AI **does not** replace physics — it **accelerates** it

Interpretation — AI

Interpretation — keywords

AlphaFold · sequence→structure · **photo** not movie · **MD** = ensemble+F · **NIM** = GPU AI · **AI+MD** · predict then sample

Git — Time Machine for Science

```
git init
git add in.lj analyse.py
git commit -m "run 1: NVT 300K"
git tag v1.0
git log --oneline
```

- **Commit** = snapshot

Git — Time Machine for Science

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Git — Time Machine for Science

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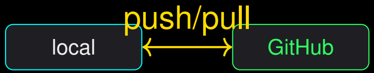
git log --oneline
```

- **Commit** = snapshot
- **Branch** = parallel try
- **Tag** = publishable version

GitHub — Lab Notebook in the Cloud

◇ **Repository**: code + scripts + `README`



reproducibility = remote backup

GitHub — Lab Notebook in the Cloud

- **Repository**: code + scripts + `README`
- **Push/pull**: backup + share

reproducibility = remote backup

GitHub — Lab Notebook in the Cloud

- **Repository**: code + scripts + `README`
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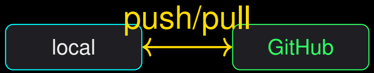
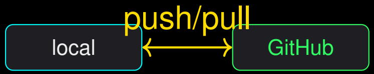
- **Issues/PRs**: discuss, review



reproducibility = remote backup

GitHub — Lab Notebook in the Cloud

- **Repository**: code + scripts + `README`
- **Push/pull**: backup + share
- **Issues/PRs**: discuss, review
- **Actions**: CI — re-run every commit



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Reproducibility — Checklist

- `README` $\rightarrow$ what + how + `uv sync`

Not Reproducible = Not Science

If someone can't re-run $\rightarrow$ it's a story, not a result.

Reproducibility — Checklist

- `README` $\rightarrow$ what + how + `uv sync`
- `pyproject.toml/uv.lock` $\rightarrow$ exact versions

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Reproducibility — Checklist

- `README` → what + how + `uv sync`
- `pyproject.toml`/`uv.lock` → exact versions
- Scripts + `lammps/build/lmp -h` + seed

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Reproducibility — Checklist

- `README` → what + how + `uv sync`
- `pyproject.toml/uv.lock` → exact versions
- Scripts + `lammps/build/lmp -h` + seed
- Data: `traj.xtc` + DOI (Zenodo)

Not Reproducible = Not Science

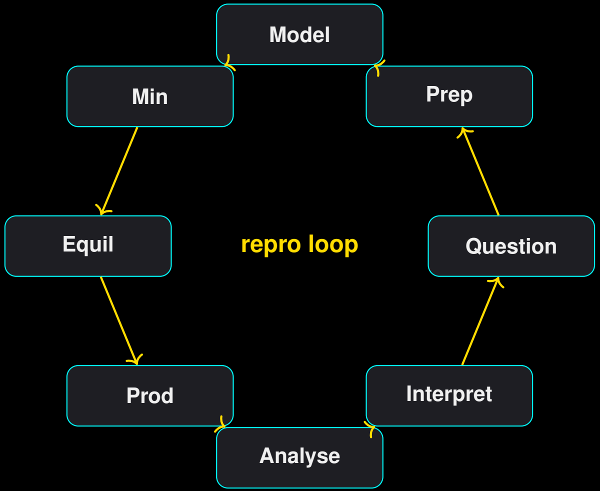
If someone can't re-run → it's a story, not a result.

Interpretation — Git/Repro

Interpretation — keywords

Git · snapshot · **branch** · try · **GitHub** · share · **README** · how · **uv.lock** · versions · **repro** or it didn't happen

The Loop — Question → Answer → New Question



System Preparation & Model

- **Prep**: crystal/PDB $\rightarrow$ `pdb2gmx` $\rightarrow$ solvate $\rightarrow$ ions

FF + Box + PBC = Lab

Your choices **are** the experiment.

System Preparation & Model

- **Prep**: crystal/PDB → `pdb2gmx` → solvate → ions
- **Model**: choose FF (**parameterization** = fit)

FF + Box + PBC = Lab

Your choices **are** the experiment.

System Preparation & Model

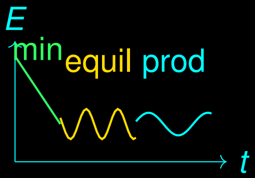
- **Prep**: crystal/PDB $\rightarrow$ `pdb2gmx` $\rightarrow$ solvate $\rightarrow$ ions
- **Model**: choose FF (**parameterization** = fit)
- **Rigorous**: validate density, $g(r)$ before production

FF + Box + PBC = Lab

Your choices **are** the experiment.

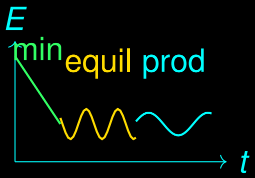
Minimization → Equilibration → Production

◇ **Min**: remove clashes, $F_{\max} \downarrow$



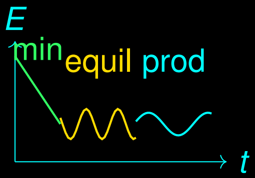
Minimization → Equilibration → Production

- **Min**: remove clashes, $F_{\mathrm{max}} \downarrow$
- **Equil**: NVT → NPT, T, P, ρ plateau



Minimization → Equilibration → Production

- **Min**: remove clashes, $F_{max} \downarrow$
- **Equil**: NVT → NPT, T, P, ρ plateau
- **Prod**: **keep** trajectory, **log** T, E



Analysis → Visualization → Interpretation

- **Analysis**: RMSD, $g(r)$, clustering, *PMF*

Reproducibility

`git tag` + `uv.lock` + `seed` + DOI → someone can repeat 5 years later

Analysis → Visualization → Interpretation

- **Analysis**: RMSD, $g(r)$, clustering, *PMF*
- **Visualization**: OVITO/ChimeraX — **watch** first

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Analysis → Visualization → Interpretation

- **Analysis**: RMSD, $g(r)$, clustering, *PMF*
- **Visualization**: OVITO/ChimeraX — **watch** first
- **Interpretation**: numbers → **mechanism**

Reproducibility

`git tag` + `uv.lock` + `seed` + DOI → someone can repeat 5 years later

Interpretation — End-to-End

Interpretation — keywords

question · drives all · **prep** · solvate · **model** · FF · **min**$\rightarrow$**equil**$\rightarrow$**prod** · plateau · **traj** · movie · **analysis** · $g(r)$ · **repro** or non-science

Thank You — Simulate Responsibly

Code · Data · Reproducibility

git clone
https://github.com/Shuvam-Banerji-Seal/introduction-to-molecular-simulation

```
uv sync && uv run python && ./lammps/build/lmp -in in.lj
```